

Introduction

Adidas is a global leader in the sportswear and athletic apparel industry, known for its innovative products, iconic branding, and strong presence across retail and digital platforms. As the company continues to expand its reach and product lines, data-driven decision-making becomes essential to optimize performance across regions, channels, and categories.

This project presents a comprehensive sales performance analysis of Adidas using Power BI. The dataset contains transactional-level data covering product types, sales volumes, operating profit, and performance across various retailers, regions, and sales methods. The objective is to identify key trends, uncover high-performing areas, and offer strategic insights that can drive more effective marketing, inventory management, and channel optimization.

Through this analysis, stakeholders can gain a clear understanding of how Adidas products perform across different markets and how to align business strategies with actual performance data.

The dataset used for this analysis contains detailed transactional records of Adidas product sales. Each row represents a single sales entry, capturing both the financial and operational aspects of each transaction. The data spans multiple sales channels (such as online, in-store, and outlet), various product categories, and a wide range of geographical locations within the United States.

Key Fields in the Dataset includes:

- **Retailer / Retailer ID:** Identifies the business partner or sales outlet responsible for the transaction.
- **Invoice Date:** Represents the date of each transaction, useful for trend and time-series analysis.

- Region / State / City: Provides geographic context for evaluating regional performance.
- Product: Describes the product sold (e.g., Men's Street Footwear, Women's Apparel).
- Price per Unit: Indicates the selling price of each unit.
- Units Sold: Number of units sold in a transaction.
- Total Sales: Total revenue generated from the sale (calculated as price × quantity).
- Operating Profit: Profit earned from the transaction after subtracting operating costs.
- Operating Margin: Profitability ratio that shows profit as a percentage of sales.
- Sales Method: Describes how the sale was made — through online platforms, physical stores, or outlet centers.

The data structure enables deep-dives into sales performance, profitability, product trends, and channel efficiency. With this rich dataset, the analysis aims to identify top-performing regions, retailers, and product categories while uncovering potential areas for optimization and strategic focus.

Data Splitting and Preprocessing

1. Data Type Verification

- Ensured Invoice Date was set to Date format to enable time-based analysis (Year, Quarter, Month).
- All numerical fields such as Units Sold, Total Sales, and Operating Profit were converted to Decimal Number.
- Categorical fields like Retailer, Region, Product, and Sales Method were set as Text.

2. Currency Formatting

- Monetary columns including:
 - Price per Unit
 - Total Sales
 - Operating Profit
- were formatted using the Dollar (\$) currency symbol for clarity and consistency.

3. Margin Formatting

- The Operating Margin column, which initially had a mix of whole numbers and decimals, was reformatted and converted to a percentage to improve interpretability.

4. Time Intelligence Preparation

- A calendar table was generated using the Invoice Date field to support time-based filtering and analysis. Additional columns like Year, Month, and

Quarter were derived.

5. Cleaning Operations

- Removed any blank rows or duplicate records.
- Renamed columns to improve readability (e.g., changing Price per Unit to PricePerUnit).

This preprocessing phase ensured that the dataset was clean, reliable, and ready for interactive dashboarding, trend analysis, and insight generation.

Pre Analysis

Story of the Data

Adidas sells products through various retailers across different regions using multiple sales methods. This dataset captures detailed sales transactions, helping us understand where, how, and what products perform best.

Industry Type of Data

The Adidas sales dataset falls under the Retail and Consumer Goods industry, specifically within the Apparel and Footwear segment of the sportswear market.

It deals with:

- Product sales performance across various categories,
- Multiple retail channels (online, in-store, and outlet),
- And geographical trends by region, state, and city.

This type of data is typically used in retail supply chain optimization, sales performance analysis, profitability assessment, and brand strategy planning. It supports use cases such as:

- Identifying high-performing markets,
- Tracking retailer and channel effectiveness,
- And improving product assortment and pricing strategy.

In essence, this dataset provides a 360-degree view of how a global brand like Adidas performs across various customer touchpoints, making it highly relevant to stakeholders in retail operations, sales strategy, finance, and marketing.

Stakeholders of the project

This Adidas sales analysis project serves a diverse group of internal stakeholders across strategic, operational, and analytical functions. Each stakeholder group can leverage insights from the dashboard to support more informed decision-making and performance improvement.

1. Sales Managers

- Track total sales, units sold, and performance across retailers.
- Identify top-selling products and high-performing regions.
- Adjust sales targets based on historical performance.

2. Retail Operations Team

- Monitor how different sales methods (online, in-store, outlet) contribute to revenue and profit.
- Identify underperforming locations or retail partners.
- Coordinate inventory allocation based on demand patterns.

3. Marketing Team

- Use seasonal and regional trends to tailor marketing campaigns.
- Focus promotions on high-margin or top-selling products.
- Segment audience targeting based on regional performance.

4. Financial Analysts & Executives

- Analyze operating profit and margin across channels and regions.
- Make data-driven decisions on pricing, discount strategies, and cost control.
- Support forecasting and budgeting efforts.

5. E-commerce and Digital Strategy Team

- Evaluate the profitability of online sales channels.

- Optimize digital marketing spend based on performance metrics.
- Align inventory and product assortment with online demand trends.

6. Regional and Territory Managers

- Benchmark performance by region, state, and city.
- Identify opportunities for regional growth or restructuring.
- Prioritize partnerships and store rollouts in profitable zones.

Each stakeholder group benefits from different dashboard views, filters, and KPIs, making the analysis both comprehensive and actionable across Adidas' business ecosystem.

Potential Analysis

Which retailer contributes the most revenue or profit?

Which region/city/state is most profitable?

What are the top-selling products by units or revenue?

Do online sales generate better margins than in-store?

Are there any seasonal patterns in sales?

Who are the top-performing retailers?

Limitations within the dataset

While the dataset provides valuable insights into Adidas' sales performance, several limitations may affect the depth and accuracy of the analysis:

2. No Customer-Level Data

- The dataset does not include information about individual customers, such as age, gender, or buying behavior. This limits customer segmentation and personalization analysis.

3. No Cost of Goods Sold (COGS)

- While Operating Profit is included, the dataset does not provide COGS or detailed expense breakdowns, which would allow for deeper profitability analysis and margin validation.

4. Limited Product Attributes

- Only general product categories (e.g., Men's Footwear, Women's Apparel) are provided. More granular data such as SKU, color, size, or release type would allow for better inventory and trend analysis.

5. Missing Marketing or Promotion Information

- There is no data on marketing efforts, discounts, or campaigns. As a result, it's not possible to assess the impact of promotions on sales and profit.

6. Geographic Scope Limited to U.S.

- The dataset appears to be limited to cities and states within the U.S. A global view would provide more comprehensive insights for an international brand like Adidas.

7. No Inventory or Stock Data

- Sales and profit are included, but there's no visibility into inventory levels, stockouts, or replenishment cycles—factors that can significantly affect sales outcomes.

Project Split

Metrics

Price per Unit

Units Sold

Total Sales

Operating Profit

Operating Margin

Dimensions

Retailer

Retailer ID

Invoice Date

Region

State

City

Product

Sales Method

Post Analysis

Following the data preparation and dashboard development, a detailed analysis was conducted to uncover performance trends across retailers, products, regions, and sales channels. The insights derived from the Power BI visualizations revealed several critical patterns in Adidas' sales and profitability performance.

Retailer Performance

- West Gear emerged as the top-performing retailer in terms of operating profit, followed closely by Foot Locker and Sports Direct.
- These retailers not only generated high sales volumes but also maintained strong profit margins, making them strategic partners for Adidas.

Product Category Insights

- Men's Street Footwear recorded the highest total sales, indicating strong consumer demand.
- Women's Apparel and Men's Athletic Footwear also contributed significantly to total revenue and profit.
- Some product categories with high unit sales (e.g., Outlet stock) contributed less to profit, suggesting potential pricing or cost structure issues.

Regional and Geographic Trends

- The West region, particularly California and Texas, led in total operating profit.
- The Midwest region contributed the least, revealing a potential underperforming market.
- Geographic concentration of high-performing locations suggests an opportunity to replicate success strategies in lagging regions.

Sales Method

In-store sales contributed the highest overall profit and sales.

- Online sales had fewer units sold than outlets but returned higher profit, indicating greater profit efficiency.
- Outlets had the highest unit sales but the lowest profit, pointing to lower margins or heavier discounts.

Yearly Profit Distribution

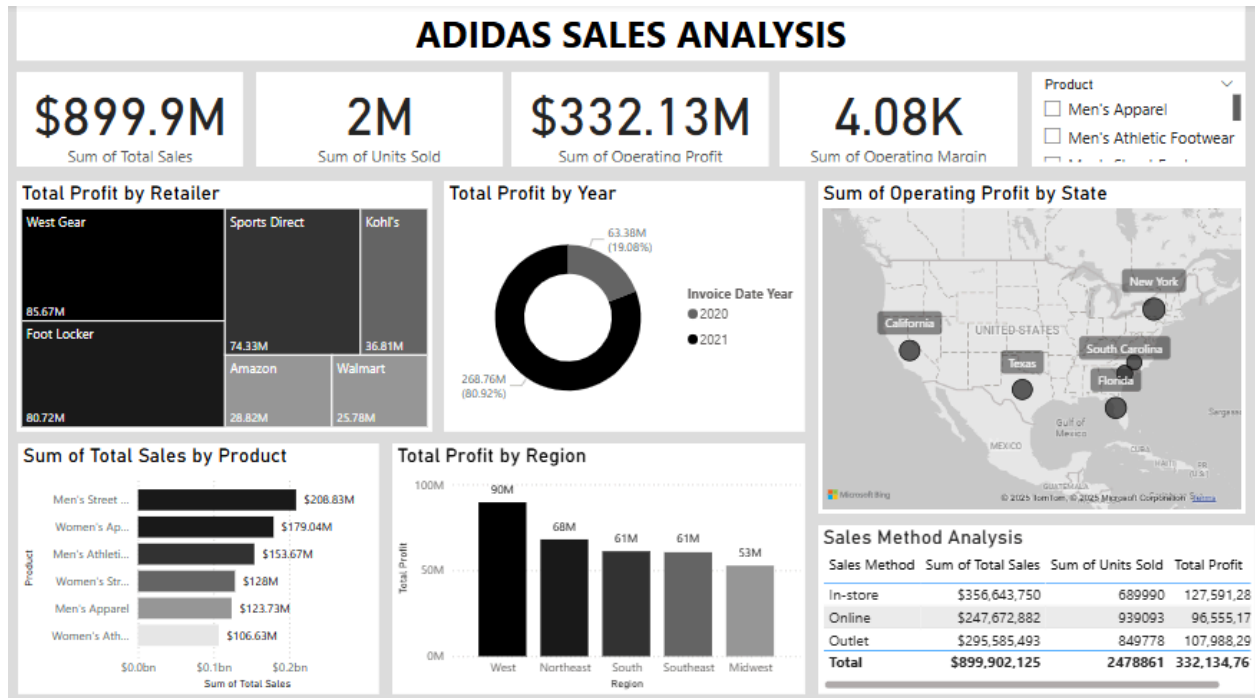
- 2021 was the most profitable year, contributing significantly more to total operating profit than 2020.
- This trend suggests either a recovery from prior market disruption or better strategic execution in recent periods.

Operating Margin Performance

- Operating Margin, once properly formatted as a percentage, showed healthy profitability across major channels and product lines.

- Some product categories and retailers, despite high sales, showed lower margin rates, indicating a need for cost review or pricing adjustments.

Adidas Sales Analysis



The Adidas sales analysis was visualized using a well-structured, interactive Power BI dashboard. The visuals were designed not just to display raw data but to help stakeholders quickly grasp performance patterns, identify key drivers, and explore the data dynamically.

Dashboard Design Focus Areas:

- Simplicity and clarity in layout
- Business-friendly metrics and filters
- Region, retailer, product, and time-based performance views

- Immediate access to KPIs and comparisons

Key Visual Elements Used

1. KPI Cards

Displayed key performance metrics prominently at the top of the dashboard:

- Total Sales (e.g., \$899.9M)
- Total Units Sold (e.g., 2 million)
- Operating Profit (e.g., \$332.13M)
- Operating Margin (e.g., 36.9%)

These indicators provide a quick, high-level overview of Adidas' performance.

2. Tree Map – Profit by Retailer

- Size and color show each retailer's contribution to operating profit.
- Helped identify West Gear, Foot Locker, and Sports Direct as the top contributors.

3. Bar Chart – Product Performance

- Visualized total sales by product category.
- Showed Men's Street Footwear as the highest-selling category.
- Supported decisions on product-level strategy.

4. Map Visualization – State-Level Profit

- Displayed operating profit by state across the U.S.
- Highlighted key markets like California, Texas, and New York.
- Made it easy to identify high-potential and low-performing areas geographically.

5. Donut Chart – Profit by Year

- Compared profit between 2020 and 2021.
- Clearly showed 2021 as the stronger year.
- Useful for spotting growth or seasonal patterns.

6. Stacked Column Chart – Profit by Region

- Compared to regional profitability.
- Revealed the West region as the top performer, with the Midwest trailing behind.
- Helped guide regional strategy recommendations.

7. Table – Sales Method Summary

Displayed key metrics (total sales, units sold, operating profit) side by side for:

- Online
- In-store
- Outlet

Allowed quick comparison of performance across channels and highlighted online sales efficiency.

8. Slicers

Interactive slicers were added for:

- Year
- Product
- Sales Method
- Retailer
- Region

These allow users to filter the visuals and perform dynamic drill-down analysis based on any combination of criteria.

Together, these visual elements create a powerful dashboard that empowers Adidas' stakeholders to explore data quickly, spot trends, and make informed decisions with ease.

Final Observations

- Retail Strength: Retailers like West Gear and Foot Locker consistently deliver high operating profit, making them valuable strategic partners for Adidas.
- Product Leaders: Men's Street Footwear emerged as the most successful product line, with strong contributions from Women's Apparel, indicating the power of targeted inventory and marketing strategies.

- **Channel Dynamics:** While in-store sales lead in volume and profit, the online channel stands out for its high profit-per-unit efficiency, indicating strong potential for digital expansion.
- **Geographic Performance:** The West region, especially California and Texas, shows outstanding profitability, whereas the Midwest region presents opportunities for strategic growth or revitalization.
- **Data Usability:** Clean formatting, consistent KPIs, and dynamic filters enabled business users to interact with the data meaningfully, fostering a culture of data-driven decisions.

Recommendations

1. Prioritize High-Performing Retail Partners

Retailers such as West Gear, Foot Locker, and Sports Direct demonstrated consistently high operating profit. Adidas should:

- Deepen partnerships with top retailers through exclusive products, joint campaigns, and incentive programs.
- Reallocate support resources from underperforming partners to high-value ones.

2. Promote Best-Selling Product Categories

Products like Men's Street Footwear and Women's Apparel generated significant sales and profit.

- Focus marketing and inventory investment on these categories.
- Expand high-performing product lines into new regions or online-exclusive campaigns.

♦ 3. Scale Online Channel Efficiency

The online sales method, while having fewer units sold, delivered strong profit per unit.

- Invest more in digital platforms, e-commerce UX, and targeted online ads.
- Explore direct-to-consumer (DTC) strategies to improve margins and customer engagement.

4. Optimize Outlet Channel Strategy

The outlet channel recorded the highest unit sales but the lowest operating profit.

- Reevaluate pricing and discounting policies.
- Consider offering curated or premium outlet collections to boost margins.
- Analyze outlet-specific costs to ensure sustainability.

5. Focus Expansion in High-Performing Regions

States such as California, Texas, and the broader West region drove high operating profit.

- Launch new campaigns, product drops, or regional collaborations in these markets.
- Use these regions as benchmarks for replicating success in similar markets.

6. Revise Strategy for Underperforming Regions

The Midwest region consistently showed lower returns.

- Conduct localized research to identify causes (e.g., low demand, competition, weak retail presence).
- Adjust marketing messages, product offerings, or pricing to better fit the regional audience.

7. Standardize Metric Formatting

The Operating Margin initially appeared as a whole number (e.g., 4.08K), which was misleading.

- Ensure all percentage-based KPIs are formatted consistently.
- Provide tooltips or explanations for financial terms in the dashboard to support non-technical users.

These recommendations offer a roadmap to improve Adidas' sales strategy through data-backed actions, ensuring smarter decisions across products, regions, and retail channels.

Conclusion

This analysis of Adidas sales data has revealed meaningful insights into the brand's performance across products, sales channels, and regions. By leveraging Power BI for data transformation and interactive visualization, we were able to convert raw transactional records into actionable intelligence that supports strategic decision-making.

The findings point to clear areas of strength—such as high-performing retailers, profitable product categories, and top-performing states like California and Texas. At the same time, the analysis exposed areas for improvement, particularly in the outlet sales strategy and underperforming regions like the Midwest.

The dashboard not only helped highlight where Adidas is succeeding but also uncovered untapped opportunities for growth. From optimizing the online channel to refining regional strategies and enhancing product promotion, the data provides a strong foundation for informed planning and execution.